Planning for Natural Language Failures with the AI Playbook

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Prototyping AI user experiences is a challenging prospect. Through 21 interviews with 12 creators of natural language (NL)-based AI products at a large technology company, we identify challenges practitioners face developing AI user experiences. One notable challenge is that of anticipating the types of mistakes that an AI will make, rendering it difficult to effectively develop and evaluate early-stage prototypes in realistic scenarios. These challenges informed the design of AI Playbook, an interactive tool for systematically navigating the complex space of human-AI interactions with common NL-based products and providing contextually relevant information for prototyping a wide range of realistic scenarios and experiences. Our evaluation of the AI Playbook demonstrates its potential to 1) encourage product teams to prioritize both ideal and failures scenarios, 2) standardize the articulation of AI failures from a user experience perspective, and 3) act as a boundary object between user experience designers, data scientists, and engineers.

CCS Concepts: • **Human-centered computing → Interface design prototyping**.

Additional Key Words and Phrases: human-AI interaction, natural language, prototyping, AI failures

ACM Reference Format:
Matthew K. Hong, Adam Fourney, Derek DeBellis, and Saleema Amershi. 2021. Planning for Natural Language Failures with the AI Playbook. In *CHI Conference on Human Factors in Computing Systems (CHI '21), May 8–13, 2021, Yokohama, Japan.* ACM, New York, NY, USA, 16 pages. https://doi.org/10.1145/1122445.1122456

1 INTRODUCTION

Software products are increasingly incorporating features that are driven by AI and machined-learned models. In many cases, designers and user experience practitioners play important roles in developing these intelligent features and experiences. However, it is not always easy to incorporate machine-learned models into established design practices. This is especially true when it comes to low-fidelity prototyping, in part because AI models are probabilistic, and can respond or fail in multitude of unanticipated ways [26]. Here, even established tools like Wizard-of-Oz (WoZ) simulations are of limited use in cases where error conditions are not anticipated, or cannot be convincingly simulated [22]. This raises the important question: How can we support product teams to prototype and test AI products early on, perhaps even before the AI model is built?

In this paper we answer this question by describing how practitioners of a large technology company currently prototype AI features, together with the challenges they face. We further present a tool, the AI Playbook, to help address these challenges. These endeavors were undertaken in three parts. First, we conducted 12 semi-structured interviews with practitioners. To scope the investigation, we focused on individuals involved in building experience that were powered by natural language processing. Participants indicated that a lack of tooling, and a focus on ideal rather than erroneous scenarios, hindered their prototyping practices. Based on these findings, we advanced to the second and

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Manuscript submitted to ACM

third phases of our work, where we developed a taxonomy of the error scenarios we felt designers should consider, and then operationalized this guidance in the form of the AI Playbook tool. The AI Playbook is an interactive tool that provides contextually relevant and actionable knowledge for simulating various language-based AI failure scenarios without any prerequisite knowledge of technical concepts in AI.

Our qualitative evaluation of the AI Playbook sheds light on its potential to 1) encourage product teams to prioritize both ideal and failure scenarios, 2) standardize the articulation of AI failures from a user experience perspective, and 3) act as a boundary object between AI designers and engineers.

In summary, this paper contributes the following:

- Delineation of practical challenges that product teams face in prototyping intelligent language-based software products.
- Taxonomy of AI failures based on a systematic characterization of 16 AI failure scenarios in common natural language-based products.
- Design and evaluation of the AI Playbook tool to support early stage prototyping of AI user experiences.

We hope that HCI practitioners and researchers can benefit from an enhanced articulation of AI errors and save time and resources in developing intelligent systems.

2 RELATED WORK

We situate our research within related work on prototyping natural language technologies, and prior categorizations of AI errors. We first establish a grounding by providing an overview of more general design guidance for AI-infused interfaces.

2.1 Design Guidance for AI-Infused Interfaces

A large body of work in HCI has culminated in several proposed guidelines and recommendations for designing AI-infused interfaces. Many of these guidelines are described at a high level of abstraction, with each addressing slightly different topics or problems in designing AI systems [1, 10, 14]. Horvitz's [14] work on principles of mixed initiative systems, for instance, provides guidance that focuses on critical factors for integrating automated services with direct manipulation interfaces. The People + AI (PAIR) guidebook provides guidance with respect to different phases of the product development process, such as identifying user and data collection needs and diagnosing errors [10]. The Human-AI guidelines provides guidance throughout different phases of user interaction (e.g., when wrong, over time, etc.) with the system [1].

Indeed, these principles recognize the need for designs to provide means to address errors when they occur. For example, the Human-AI interaction guidelines [1] provide five separate guidelines regarding the mitigation of errors, including: supporting efficient dismissal and correction, and gracefully degrading the AI's proactive capabilities when uncertain about a user's goals. Likewise, the PAIR [10] design principles focus on aspects of managing user expectations of uncertainty in AI errors, such as defining and identifying errors and their sources and providing paths forward from failure. While these principles provide high-level guidance on how to efficiently prevent and recover from errors, they often lack low-level concrete advice that is both immediately actionable, and grounded in a specific application context.

Our paper offers a complementary approach by facilitating designers in simulating the specific types of errors their users are likely to encounter. Designers can then verify that their proposed designs adhere to the guidelines and principles, across a broad and realistic range of error conditions.

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2.2 Prototyping AI Systems

To support rapid prototyping and testing of AI interface designs, there is a large need to investigate methodologies that enable designers to simulate AI system characteristics at various stages of development.

2.2.1 Medium-fidelity prototyping. Mid-fidelity prototyping techniques allow users to define limited model behavior to approximate a finite range of system functionalities before a final model is available. For example, systems such as Crayons [8] and the $1 gesture recognizer [24] enable users to quickly approximate a final AI's behavior by presenting them with easy-to-configure models. Likewise, crowdsourcing platforms and techniques allow researchers to prototype partially automated (or humans-in-the-loop) systems by constructing a computational pipeline in which micro-tasks are distributed among crowdsource workers. Humans-in-the-loop computing allow researchers to prototype and deploy realistic natural language systems at scale, such as email-based meeting scheduling system [5] or intelligent speech-based system for prototyping user interfaces [16].

In both cases, these tools require partial implementations and functional code to enable the interactions. Such systems can be costly to produce early-on. In this paper we focus on the earliest stages of ideation and prototyping.

2.2.2 Low-fidelity prototyping, and Wizard-of-Oz simulation. Low-fidelity prototyping techniques such as sketching and paper prototyping allow designers to rapidly explore multiple alternative design choices so as to *get the design right* without significant investment and implementation overhead [3]. Testing these prototypes however require significant interventions by human experimenters. To this end, Wizard of Oz (WoZ) prototyping was originally conceived and developed in the context of envisioning future scenarios for intelligent language experiences, and often involved training the human experimenter (or the 'Wizard') to act more like a computer [6, 9, 11, 15]. Over the course of three decades, HCI researchers continued to adapt and refine the WoZ technique by adding mechanisms to automate individual components of the WoZ protocol to reduce latency, ensure consistency, and better simulate realistic system responses [21, 22]. For example, panels of wizards can be leveraged to distribute the burden among multiple people, thereby reducing latency.

Still, prototyping intelligent behaviors with WoZ techniques remains challenging due to difficulties in completely eliminating delays caused by aspects of protocol, or simulating consistent system behavior and probabilistic errors, ultimately making it difficult for participants to believe they are interacting with an intelligent system [22, 26]. Only recently Yang et al. proposed promising approaches to simulate NLP errors by designing a hybrid of WoZ and off-the-shelf ML toolkits to best simulate likely errors [25]. For example, they suggested using rounds of machine translation services in order to simulate 'noise' in context detection, by translating an author's written English sentence into a foreign language and back to English.

In our studies, we observed prototyping challenges similar to those encountered when developing convincing WoZ simulation; namely, that of producing realistic AI behaviours and errors for consideration. In the next section, we describe work categorizing the types of AI errors that we consider in our research.

2.3 Types of AI Errors

Errors are an unfortunate byproduct of AI systems that rely on imperfect machine learning models. Prior research provides a broad catalogue of errors in AI and natural language processing systems. One approach is to identify the source of errors from language-processing models. For example, Moldovan et al. analyzed modules of a linear NL model used in an open-domain question answering system to characterize information retrieval errors [19]. The

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authors identified 10 modules, or sources of errors, ranging from keyword preprocessing (e.g., spellcheck) to response formulation (e.g., answer string shifted slightly to left or right).

Other researchers analyzed system generated errors and relied on theoretical framing to characterize them [2, 7, 12]. According to Herbert Clark's grounding model of communication, participants in conversation coordinate on four different levels to achieve mutual understanding, including channel, signal, intention, and conversation levels [4]. While intended for human-human communication, Paek and Horvitz made the case to use the same coordination scheme for grounding human-machine interaction [20]. Here, errors can be classified into four categories depending on which level the errors occurred. For instance, channel level errors occur when a speech-based AI system fails to properly attend to user's attempts to open a channel of communication (e.g., uttering a wake command). Signal level errors occur when the system fails to process the user's spoken utterance due to errors in transcription. Intention level errors happen when the system fails to comprehend the semantic content of the user's spoken utterance (e.g., calendar application fails to recognize 'erase event' instead of 'delete event'). Finally, conversation level errors occur when the user formulates a query that falls outside of the system's capabilities (e.g., asking calendar application about weather).

Our taxonomy of AI failures derive extensively from Clark's grounding model, which we use to characterize 16 AI failure scenarios that occur in common NL-based AI products. To our knowledge, we contribute the first systematic characterization of AI failures that affect the end user experience of NL products.

In the remainder of this paper, we describe our process for navigating the complex challenges that practitioners face in prototyping, and unpack how an interactive tool like AI Playbook can deliver promising approaches to support UX practitioners in planning for natural language failures.

3 EXISTING AI PROTOTYPING PRACTICES

To understand how to support AI interaction designers, we first needed to understand their existing processes, and the practical challenges they face in prototyping and testing AI systems. To do this, we conducted semi-structured interviews with product teams in a large technology company.

3.1 Interview Study

3.1.1 Recruitment. We recruited 12 practitioners through a snowball sampling strategy. Our inclusion criteria included product teams working with NL technologies who were willing to participate in a follow up user study (described in Section 5 of this paper). Our participants had one to 15 years of experience in practice and include a mix of user researchers, designers, and program managers—representing various language intelligence product teams working in areas of search, writing assistance, conversational AI, and recommendation (See Table 1). IRB approval was obtained prior to the consent and data collection period of April 9 to June 2.

3.1.2 Interview. Due to the COVID-19 pandemic, we used a video conferencing tool to conduct the study. Interview topics included practitioner challenges spanning the AI design and development process, team collaboration, cost of AI failures, and evaluation. As an example, we asked the following questions to elicit practitioner experiences related to these topics: what does the prototyping process look like? What does collaboration with a data scientist look like? When does the jump between code and design happen? Tell me if there were any surprises when you deployed the feature. All interviews lasted no more than 30 minutes, were audio recorded, and transcribed for analysis.

3.1.3 Analysis. We collected 352 minutes of audio data, and thematically analyzed the transcripts through multiple iterative phases. The first author coded the transcripts through an open coding process, identifying and noting text

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Table 1. Demography of study participants. *=participation in first study only; Exp=Experience in years of practice

ID	Practitioner Role	Product Team	Exp.	Envisioned Feature in User Study (Task II)
P1*	User Researcher	Writing Assistance	1	-
P2	Program Manager	Writing Assistance	10	Text prediction in writing assistance
P3	Designer	Writing Assistance	5	Tone detection in writing assistance
P4	Design Manager	Conversational AI	10	Content consumption in conversational AI
P5*	Program Manager	Search	9	-
P6	Design Manager	Search	4	Recommended content in in-app search
P7	Program Manager	Search	15	Query completion in in-app search
P8*	Program Manager	Speech & Search	3	-
P9	Program Manager	Recommendation	15	Recommended designs in presentation app
P10	User Researcher	Recommendation	5	Query completion in in-app search
P11	User Researcher	Conversational AI	1	Recommended content in conversational AI
P12	User Researcher	Writing Assistance	2	Text detection in email communication

segments related to challenges in prototyping and testing AI experiences. In a second pass, the author organized the annotated quotes into multiple low-level subthemes. The first, second, and last authors then gathered in multiple meetings to review the annotated quotes and assess thematic relevance between subthemes, arriving at major themes which provide coverage for identified clusters. Through successive meetings, all authors discussed and refined the themes until no new themes emerged.

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3.2 Findings

We initially set out to examine product teams' practices and challenges with respect to prototyping. To our surprise, however, early prototyping was rarely considered in their practice. Moreover, when a prototype was developed, numerous practices, challenges, and compromises served to limit the validity of any insights derived from the prototype or its evaluation. Below we unpack key forces that give rise to these issues.

3.2.1 Prototyping slows development. In several cases participants explained that, in considering the often tight product timelines, traditional prototyping and testing methods were too costly to perform—slowing down the otherwise fast pace of development. For example, P4 said "*a lot of prototyping that we do is, it takes a lot longer than we typically have.*" In another example, P6 said "*I think it's a little bit different within [search product], because things happen so fast that we tend to skip over prototypes. We do prototyping for really, really big things, but a lot of we do, it just goes straight to code that gets flighted.*" (i.e., limited audience A/B testing of a feature in an already shipped product)

3.2.2 Lack of tool support. The slow speed of early-stage prototyping was often attributed to a lack of established methods and tools. As noted above, participants expressed a preference for 'flighting', but flights require a shipped product, with an established audience, and a robust instrumentation pipeline. In cases where flighting was not possible, practitioners felt they lacked comparable tooling for evaluating prototypes. For example, P3 remarked, "*I don't know how you prototype that. Right? The equivalent for prototyping was to build the options [intelligent features] and then flight them.*" Likewise—and again noting the fast pace of development—P4 explained, "*this is a super ambiguous space. The most ambiguous space that I've ever worked in, in my years of working in design. We are literally making this up every day as we go along because things are moving so fast... there aren't any real rules. And we don't have a lot of the tools.*"

3.2.3 Hero scenarios and golden paths. In cases where prototypes were developed, participants noted several threats to the validity of any insights derived from the development or evaluation of the prototype. In particular, numerous participants discussed their team's focus on high-value 'hero' scenarios, which were often conceptualized, specified, and prototyped under ideal circumstances with ideal data. This made it easy to overlook possible AI failures and error scenarios that deviate from this 'golden path'. For example, a PM in the writing assistance team remarked "*we focus more on hero kind of scenarios.*" Similarly, P7 said "*I think design still oftentimes is in the space of designing for kind of ideal solutions.*"

P8 described their experience releasing a hero experience and then only later addressing a known user experience issue: "*for the MVP*[1] *that we shipped two years ago, we ignored the problem of the speech engine returning low confidence results. For the design which is just shipping now, we actually came up with a system that said, 'Well, when we get low confidence results back, we're going to provide a visual treatment to [the user]'.*"

Failure to identify and address errors early incurs technical depth—it can be costly to discover and address such errors later in development. Indeed, several of our participants described experiencing such costs. P3 noted, for example, that it became more difficult to collect rich qualitative feedback post deployment: "*[users] only react when there is something super positive or super negative that's happening to them*" while P1 emphasized the significant burden of reviewing and gaining meaningful insights from that feedback: "*how many of these results are we going to get a day? How are we going to parse through them and understand? It's a lot of burden on both ends.*" When issues are identified from post-deployment feedback, the prohibitive costs of addressing those issues can lead teams to reactively address only the most severe cases "*unless it's like a super severe escalation... it's going to be hard for us to flag it*" (P1). Moreover, the difficulty of updating an already deployed system often results in a slow turnaround time in improving the user experience "*if I do design today, I won't actually be able to see [user feedback] in an internal [deployment] in a product for almost three months. So, my ability to effect something is really delayed*" (P4).

3.2.4 It is difficult to anticipate errors. Unfortunately, when teams did attempt to identify potential errors and failure scenarios during early stage design and development, many participants described difficulties anticipating such failures, or approaching failures in a systematic way. For example, P1 said "*it's so hard to know until the bad thing happens.*" Similarly, P9 described the difficulty accounting for the many ways an AI system might fail: "*sometimes we may be able to identify a result that we're highly confident in. And in those cases we try to go for design... But we also need to account for the many, many times where we have very little context to go on where we may have very low confidence.*" Another participant, P5, described a trial-and-error approach to understanding errors early in design, saying "*we started mucking around a little bit, rerunning things and then switched out some parameters. So, a little bit of trial and error.*"

3.2.5 It is difficult to prototype personalized experiences. A second threat to the validity of the insights derived from prototypes arises from the challenges of simulating personalized or highly-contextual experiences. For example, P9 said "*that's not something that we normally do...[It] takes a lot of time and it's very personal. Like it's one thing to say, 'hey, come down to [company] and we'll pay you 50 bucks for your time'. And just, you know, use a computer for us. It's another thing to say, 'and by the way, bring your data along'. So that changes the conversation quite significantly. So that's why we don't do that.*" Similarly, P1 said "*well, we can't learn about how people are going to react to it [AI writing assistant] until we ship it.*"

[1] Minimum viable product

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In the few cases where participants did describe efforts towards early prototyping of personalized experiences, they emphasized the limitations of current techniques for doing so. For example, some participants described using synthetic data (e.g., pre-canned text, email accounts, etc.) or simple rule-based models to simulate planned models for early testing. However, these participants also lamented that neither approach delivered AI experiences that came close to what users would experience in the context of their personal data. For example, P1 said "*we simulated [the intelligent feature] very early on but like with [...] fake dummy writing, but the results just didn't have the depths we were really looking for. So that's why in the end it just worked best to not simulate it at all and just test out, like even if it's buggy. At least we'll have people's real writing so people can truly assess the quality of the results and the experience of how we surface these results.*"

3.2.6 Breakdowns in interdisciplinary communications. Finally, we note the development of AI-driven interactions is inherently cross-disciplinary. However, along with the lack of tools and techniques to support efficient and effective early prototyping, several of our interviewees described challenges collaborating and communicating across interdisciplinary teams.

In several cases, participants described impediments to iterative prototyping due to late and limited communication about identified issues. For example, P6 said "*design was rarely involved in the conversations with [engineers], which led to kind of like a game of telephone, where like stuff would be happening that would impact the design over here, but nobody would tell the designers until a bunch of stuff had been done, and then [the engineers] would have to go redo it because it wasn't matching.*" In another example, P1 described feeling hesitant to raise issues late after an AI model is developed knowing the cost that might be incurred to address those issues "*oftentimes it's just, these projects come from PM of like, this thing is happening, our data scientists already have a model, let's test what this looks like in real life. Like, understanding the usability and comprehension of it. And in a way, like, it just feels bad because... I feel like some of my research like, is crapping on a lot of really great work that our data scientists have done to perfect the models.*"

Participants also described challenges engaging across disciplinary boundaries due to varying levels of familiarity with AI and ML concepts, including understanding what can go wrong and what can be easily updated. For example, P2 said "*it's sometimes challenging so, if you talk to somebody who haven't been working in the AI domain... sometimes that's a challenge to communicate that as an idea... like communicating to designers and researchers to show, like what is the problem we may face?*" Similarly, P3 remarked that "*a better understanding of what are the things that you can push back, or you can change*" would help empower team members to contribute to the design and development of AI-based products.

In summary, our participants indicated that they did not often prototype new features or experiences, in part because they lacked tooling, and felt it slowed down the pace of development. In cases where prototyping was done, a focus on golden paths and hero scenarios, together with the use of synthetic data, threatened the utility of the prototype, and the validity of user feedback. Each of these challenges was further exacerbated by breakdowns in cross-disciplinary communication.

In this paper we introduce the AI Playbook to address the above-mentioned challenges. The AI Playbook provides low-cost tooling for identifying a broad range of error scenarios in the context of a given AI experience. As we will later show, the AI Playbook can serve as a boundary object [23], bridging UX and data science disciplines by grounding AI failure modes in user-facing scenarios. In the next section we enumerate and organize these failure-modes—a necessary first step in the development of the AI Playbook.

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Table 2. Taxonomy of natural language failures. Taxonomy consists of failure types, sources and scenarios.

Failure Type	Failure Source	Failure Scenario	Example Failure Scenario with NL Products
Attention (channel level)	Missed trigger	System fails to detect a valid triggering event.	[Scheduling assistant] fails to detect a meeting request in an email. [Voice assistant] fails to detect a wake word.
	Spurious trigger	System triggers in the absence of a valid triggering event (it triggers when not intended).	[Scheduling assistant] mistakes meeting minutes as a new meeting request. [Voice assistant] mistakes background speech as a wake word.
	Delayed trigger	System detects a valid triggering event, but responds too late to be useful.	[Scheduling assistant] detects a meeting request, but offers support only after the user has already manually scheduled the meeting.
Perception (signal level)	Truncation	System begins capturing input too late, or stops capturing input too early, and thus acts only on partial input.	[Voice assistant] interprets a pause as the end of an utterance, and answers before the user has finished speaking.
	Overcapture	System begins capturing input too early, or stops capturing input too late, and thus acts on spurious data.	[Voice assistant] captures user's intended query along with a few more words spoken after the query.
	Noisy channel	User input is corrupted by spelling errors in written text, or background noise in audio input.	[Search engine] user misspells their search query. [Voice Assistant] background noise, such as music, makes speech less discernible.
	Transcription	System generates common transcription errors such as homonyms, homophones, plural word forms, etc.	[Automatic captioning service] fails to transcribe certain speech utterances by adding or removing inflected endings (e.g., -ing, -s, -ed).
Understanding (intention level)	No understanding	System fails to map the user's input to any known action or response category.	[Virtual assistant] responds "I'm sorry, I don't know how to answer that question."
	Misunderstanding	System maps the user's input to the wrong category of action or response.	[Shopping assistant] mistakes an item refund request for an item exchange request.
	Partial understanding	System correctly infers some aspects of the user's intent, (e.g., correct action or response category) but get some details wrong.	[Travel assistant] mistakes the origin for the destination. [Scheduling assistant] fails to consider a location's timezone in a meeting request.
	Ambiguity	There may be several reasonable interpretations of the user's intent, leading to ambiguity. The system fails to correctly resolve ambiguity.	[Search engine] responds to the query "US Open" by returning results for tennis, when the user intended golf. [Scheduling assistant] sends a meeting request to the wrong "John".
Response generation (conversation level)	Action execution	System fails in executing the desired action.	[Travel assistant] attempts to re-book a trip by canceling and reissuing a ticket, but fails partway, leaving the reservation in an unknown state.
	Generative response	System generates an incoherent, inappropriate, factually incorrect or partially correct response.	[Social chatbot] produces an offensive response to a user's input.
	Ranked list	System produces a ranked list with low precision, recall, or result diversity.	[Presentation design assistant] offers five recommended slide designs, but two designs are duplicates, or are otherwise indiscernible.
	Binary classification	System generates a false negative or false positive response.	[Spam filter] misclassifies an email as unsolicited spam. [Content moderation system] misclassifies a comment as obscene.
	Multi-class classification	System fails to produce correct classifications among close or distant categories.	[Email client] classifies a receipt as a promotional email.

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4 TAXONOMY OF NATURAL LANGUAGE FAILURES

We learned very quickly from our interviews that most practitioners were not even aware of what errors were possible in their application contexts. This rudimentary challenge prompted us to systematically taxonomize the types of errors that may occur in their practice. Given our participants' extreme sensitivity to anything that would delay development, we elected to focus on errors that are either very common, or rare but very costly. Moreover, we focused on errors that directly affect the end-user (as opposed to other users of the system, or other stakeholders), and can be observed in a

single session of application use (as opposed to errors that arise over time). While not exhaustive, we believe these decisions strike the right balance of delivering high-value at low cost.

To develop the taxonomy, eight computer science researchers with relevant expertise in Human-AI interaction and natural language technologies, including three authors of this paper, convened to discuss relevant AI failures. Through multiple meetings and small discussions, we developed the hierarchical structures of the taxonomy in two steps. To determine the main categories in the hierarchy, we first looked to the grounding model that was proposed by Herbert Clark [4] and later adapted by Paek and Horvitz [20]. We adapted their four constructs, including *channel level* (attempt initiate communication), *signal level* (understanding what behavior is intended as signal), *intention level* (understanding of semantics), and *conversation level* (in which a response is generated), to form the highest hierarchical layer of our taxonomy, namely: attention, input perception, understanding, and response generation errors (Table 2, leftmost column).

Next, within each category, we identified potential sources and examples of those errors, pulling from prior literature that examined natural language processing errors in the domains of information retrieval [19] and discourse and dialogue [2, 7, 12], as well as from personal experiences in developing such systems.

It is important to note that the taxonomy is designed to capture a broad range of scenarios. Not all failure types and error sources will apply to every product or feature. For example, attention/triggering scenarios are unlikely to apply in cases where the AI experience is explicitly and unambiguously invoked by the user. Likewise, generative response errors may not be relevant to systems that produce a ranked list over existing items. To this end, we evolved the taxonomy into the interactive AI Playbook tool, which leverages a survey instrument to recommend error scenarios that are relevant to a given application context. We describe this tool next.

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5 AI PLAYBOOK

Our distillation of practitioners' challenges led to high-level goals for design. Our taxonomy of natural language failures provided the necessary content and structural support to realize those goals in the form of an interactive tool. In the following sections, we describe the design and evaluation of the AI Playbook.

5.1 AI Playbook Design

A first step in the design of the AI Playbook was to translate the high-level goals into a more concrete set of design desiderata. Specifically, to integrate prototyping into our participants' real-world workflows, we require the AI Playbook to:

- Provide a means to discover non-ideal situations that are outside of the golden path scenario.
- Provide actionable and contextually relevant guidance on how to simulate these scenarios.
- Provide a means to explore the full range of options and any consequences of in-the-moment decisions.
- Provide the above-mentioned features as efficiently, and low-cost as possible.

We depict the the AI Playbook in Figure 1, and describe its key features in the sections below. The AI Playbook is also available online at https://aiplaybook.github.io/.

5.1.1 Interactive survey. Using the taxonomy as a foundation, we designed an interactive survey (Figure 1-A) that asks practitioners to walk-through their envisioned product or feature by answering questions about the type of system they are designing, input modality, trigger, delimiter, and the form of the expected response. Branching logic ensured that the survey questions, examples, actionable advice, and help, were relevant to the current context. For example, if

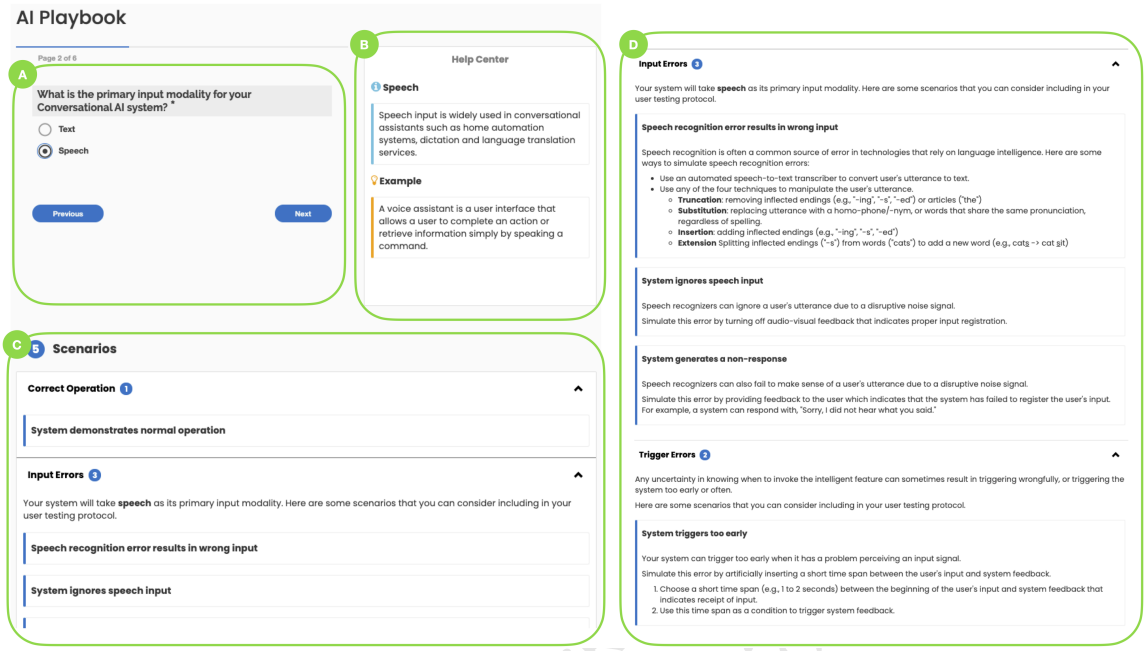

Fig. 1. Screenshot of the AI Playbook user interface. A: Interactive survey with buttons that allow users to navigate between different question and answer pairs; B: Help center displays supportive information (e.g., description and examples) tailored to the user's selected answer choice; C: Scenario builder allows users to view potential failure scenarios based on their selected answer choice; D: Playbook report allows users to view the full list of scenarios along with detailed guidance on how to simulate them.

users indicated their feature used speech as an input, subsequent questions were phrased accordingly. In early design meetings, the design team analogized this aspect of the Playbook as being "like tax preparation software".

5.1.2 Help center. To support practitioners' understanding of the concepts indicated in the questions, we designed a help center (1-B) that shows a general description of NL concepts along with specific examples from existing products that demonstrate how the concept is realized in real world scenarios. As noted above, this content was contextually relevant, and based on the user's current and prior selections.

5.1.3 Scenario builder. We designed the Scenario builder interface (1-C) to enumerate specific scenarios for the designers to mock up, prototype, or, at the very least, consider, when developing their designs. As the user progresses through the survey, scenarios accumulate, providing real-time feedback on the consequences of various survey choices and design decisions.

5.1.4 Playbook report. The scenario builder can be expanded into a detailed report (1-D). The report provides actionable guidance on how to simulate each scenario. A partial report can be accessed at any time by expanding the items in the scenario builder, and the full report is generated automatically at the completion of the survey.

5.2 User Study

After developing the AI Playbook, we presented it to the participants of our interview study for evaluation and further discussion. We describe the results of this second qualitative study phase below.

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5.2.1 Recruitment. As noted above, we invited participants from our previous interview study to comment on the AI Playbook, allowing us to evaluate if our design was meeting their needs as intended. 10 out of 12 previous interview study participants responded to an email study invitation, and nine participants consented and completed the follow-up user study.

5.2.2 Procedure. The AI Playbook was developed to run on a web browser, and the application was hosted via a local server on the facilitator's (first author) computer. Due to the COVID-19 pandemic, we used a video conferencing tool to share the facilitator's screen and gave study participants access to control the mouse cursor and keyboard to remotely interact with the AI Playbook system.

We asked participants to complete two tasks using the AI Playbook. In Task I, we asked participants to use the tool to facilitate prototyping a fictional chatbot system designed to help website visitors search for an apartment. In Task II, we asked participants to apply the tool to a feature or system that their product team was currently in the process of conceptualizing or prototyping. If participants were not currently in that phase of design, we asked them to consider the most recent real-world experience that applied. Table 1 lists the features participants considered in Task II.

Participants were encouraged to think out loud as they completed each task. A semi-structured interview was conducted at the end of Task II. Through these interviews, we elicited qualitative feedback spanning topics including: concrete aspects of the tool, the suitability and applicability of the tool to their current work practices, the perceived or anticipated value of the tool, and opportunities for further development. The entire procedure took 60 minutes.

Sessions were screen-recorded, and later transcribed. Immediately after their completion of each task, we downloaded a PDF copy of the generated reports. We also immediately sent copies of the reports to the participants for additional scrutiny, and to ground the subsequent semi-structured interviews. Data collection took place between June 22 and July 10.

5.2.3 Analysis. We obtained 479 minutes of screen and audio recorded data, which were transcribed verbatim. All qualitative analysis of transcripts was conducted in similar fashion to the prior interview study. Any specific references made with regards to the design or contents of the AI Playbook in the transcripts have been cross-referenced with relevant video footage of participants' usage. We then took notes from our observations of screen recordings and included them in the thematic analysis.

5.3 Findings

We now turn to share key insights from our evaluation of the AI Playbook. Our findings are organized such that they mirror the above-mentioned design desiderata, then delve into other aspects uncovered in the interviews.

5.3.1 Consider scenarios outside of the golden path. After their use of the AI Playbook, many participants were convinced that a tool such as AI Playbook could help them consider atypical situations or failures that are too often outside the radar of the golden path scenario. For example, P6 commented: "*I think so often we only test the perfect scenarios. Like we only design and test the end-to-end experience exactly as it is supposed to work, and I think this [AI Playbook] would actually help us to test where it might break [...]. So if something didn't trigger when it was supposed to, how much does that impact a user? Would it impact how much they want to use it in the future? Or something like that. And we could actually design and test for that instead of designing these closed loop perfect end-to-end things. This gives us some flexibility and I think it would change the process.*"

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Participants also suggested that the tool can help product teams navigate the error space systematically, instead of implementing ad hoc approaches. For example, P3, a designer, reflected on her team's struggled to generate mistakes in an arbitrary, ad hoc manner: "*this is something that we did manually at first [...] we made mistakes on purpose and wanted to figure out if people will catch on to that and how will they react.*" Likewise, P7 reflected: "*I've not had the experience of like going through an audit of the potential errors and saying 'Okay, for each one of these things, what are you going to do from a design point of view?' It oftentimes is more organic. It kind of happens ad hoc you know? Design puts red lines down. [...] Engineers realize, 'hey, we're going to run into this particular error.' We go back to design. [...] There's lots of efficiency that is lost there. So something like this can help with that efficiency.*"

In addition to helping teams systematically navigate the error cases, participants saw that the tool could help standardize the types of errors that practitioners can include in prototyping and testing. P2 shared his excitement about the potential opportunities: "*What I see would be beneficial from this is, you will standardize the error case design. You could potentially help to improve the the PM spec a lot.*" Likewise, P9 explained: "*I see it as a checklist tool to make sure that like we are able to force a consistent bar across different teams in different organizations.*"

5.3.2 Provide actionable and contextually relevant guidance. While the AI Playbook was deemed useful in stepping off the golden path, it was also found to provide actionable guidance that participants felt they could immediately apply. For instance, P4, a design manager explained: '*we deal with all this stuff everyday. But yeah, there's some good notes here about how to test and validate for these things. So yeah, very real world for us.*" Likewise, P2, a program manager, remarked: "*I think this level of details are enough so we can do a proper documentation and then guide engineers to do their work.*" Similarly, P6, also a design manager, said the details were enough to impact how she would approach writing her user study protocols: "*even if we are thinking about [error scenarios] it's not to the depth that this Playbook surfaces. [...] I would rewrite so many studies if I had in this list of scenarios.*"

Even when guidance was deemed not to be perfectly applicable, some participants felt they could easily adapt and tailor the guidance to their situation. P6 expressed this sentiment best as follows: "*to me, this serves as a really good example of what to do. It may not be the exact solution for every feature. That's fine. It gives me the right direction to come up with my own solution.*"

5.3.3 Provide a means to explore the full range of options and the consequences of decisions. Some participants also appreciated the comprehensive nature of the tool, and the ability to immediately observe the consequences of their actions. For example, P11 explained: "*I think it's a very helpful tool. Like when you think about it, it gives a very concrete way to articulate like all the ways in which the system could fail.*" Likewise, P4 noted "*the fact that it's so short is nice, because if you got to the end and the recommendations were kind of missing a piece of what you were doing you would sort of realize, 'oh, maybe I should try a different category and see if I get something that feels a little closer'.*"

Nevertheless, our design may have erred too far in favor of exploration. Some participants left wondering about the scenarios omitted from the report, and longed for comprehensive list of errors—including those deemed not to apply. For example, P10 noted an analogical reference to her own experience testing a recommendation product, "*a lot of time [users] would be like, 'this is great. It's relevant, but like I want to see everything' [...] and [the short list] is not giving them the confidence that they're finding like the best. So they needed to know like 'how relatively good is this'?*" Likewise, P3 offered careful advice that too much contextualization would not be ideal for new and inexperienced practitioners: "*I imagine that if someone is starting to work in that area, [...] they wouldn't even know where to look for the problems. They will have to go through all the process in order to say 'OK, we didn't check for that. Oh, I wish I knew!'*"

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5.3.4 Deliver features as efficiently, and low-cost as possible. After using the AI Playbook, all participants appreciated the way in which the AI Playbook's format allowed them to efficiently cover a lot of ground in three to five minutes of their time. P4 expressed this sentiment best:"*You know, that's pretty low investment, right? For a high [value] trade off. Yeah, I really enjoyed this.*" Some participants attributed the high efficiency to the tool's ability to support their easy navigation and exploration. For example, P6 expressed her enthusiasm for the 'help center' feature as follows: "*This, I mean, that help center was my favorite part of this whole process. [...] I'm sure you saw me change my answers a bunch of times as I would read the different things because it helped me [...] I really liked the examples in there too, because that's what actually helped me the most.*"

5.3.5 Bridge considerations across teams and disciplines. While we did not explicitly include support for team communication and collaboration in our desiderata, many saw the potential value that the AI Playbook can serve in bridging the culture and language of practice. For example, P6 remarked: "*we all feel like we're sometimes speaking different languages, and this [AI Playbook] just instantly like levels it out and says, here's your report [...] I feel like it puts everybody on the same level in terms of being able to use the same terminology to have very clear scenarios laid out.*" More simply, P10 noted "*I would use [the Playbook] as a tool to connect the disciplines around these issues*", and P3 added the the AI Playbook served as a "*kind of a boundary object*" between interdisciplinary teams.

Going further, some participants emphasized the value of the AI Playbook in encouraging shared accountability and consensus building. P6 in particular emphasized this point, saying "*I actually feel it [...] could be a great exercise for everyone on the team to do this individually and then we could compare how we all answered it. I think that also then identifies where we may not be on the same page.*" Likewise, P9 suggested a potential for the tools to drive accountability, perhaps automatically scheduling review meetings. He explained, "*I think the one thing to really be mindful is to not let this be a tool where people just go through the motions and have a result and then just kind of move on with their lives [...] So at the very least, there's some sort of accountability in making sure that the users of the system actually go through it and actually think about it.*" P4 also agreed with this sentiment and suggested making use of existing group meetings such as brown bag events or workshops to reinforce product teams' group knowledge of the error scenarios. "*not just dump it in email and say 'check it out', but more use it as one of our brown bags or one of our team hands-on workshops or something to get people more engaged in the conversation.*"

Finally, an important value that participants saw in their use of the AI Playbook is its potential to empower UX practitioners to communicate their opinions more effectively with interdisciplinary teams. P6 shared her excitement, "*it felt like, 'oh, I could actually go and have a conversation with a ML person about these issues' and feel confident going into that—which I think would just build better team dynamics.*" Likewise P11, expressed a similar sentiment, and considered the tool as an added resource when communicating with designers. She said, "*I could see this as if I'm having a conversation with the designers who are actually building this prototype and saying. Hey um. You know, like 'what happens if we're making bad recommendations' or 'I think we should include some scenarios that maybe don't [succeed]'.*"

6 DISCUSSION AND FUTURE WORK

In the following sections, we reflect on limitations of this research and point to future opportunities for further discussion and investigation.

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6.1 Generalizing Our Findings to Companies, and to Industry at Large

While we carried out this research project in a single large software company (comprising numerous disparate teams and organizations), we recognize that our findings cannot directly speak to the practices of other companies. However, people in our industry are highly mobile, often transitioning between teams and companies, and bringing their practices with them. To this end, we hypothesize that similar practices are observed elsewhere in the industry. To confirm or reject this hypothesis, and to better understand cross-industry practices, we welcome future work in this space.

6.2 Extending to Other AI Scenarios

The AI Playbook covered a limited set of AI failure scenarios, focusing on natural language-based products, yet we imagine the same approach could be used for a wide variety of AI-based systems (e.g., scenarios that utilize computer vision).

Likewise, the AI Playbook does not consider AI failure types that are contingent on the end-user's interpersonal, sociocultural, and societal context; nor does it consider systematic errors that result in unethical and unfair outcomes such as favoring one user group over another. Indeed, our participants talked about extending the AI Playbook to cover such scenarios, and even provided examples of real-world situations in that resulted in racially or socially insensitive outcomes (e.g., suggesting the response "Great!" to a sad message about a cancer diagnosis). Despite the importance of such errors, a full understanding of this space is an active area of research [13, 17, 18], and we welcome expanding the AI Playbook to better predict in which circumstances such errors may arise, and to provide better guidance on how they can be simulated and ultimately mitigated.

The AI Playbook also does not consider errors that manifest longitudinally or via personalization. As noted in Section 3, personalized experiences inevitably require access to users' personal data, yet such access invites other technical and ethical considerations around privacy.

Finally, the AI Playbook considers only scenarios in which the end-user is directly impacted by an error. In other scenarios, such as adversarial attacks on ML models, it is important to consider a broader range of stakeholders.

In considering each of the above-mentioned cases, we note that the platform is extensible, and that teams can adapt and extend the Playbook to accommodate such scenarios.

6.3 Standardizing Terminology

Though participants noted that the AI Playbook had the potential to standardize communication across teams and disciplines, one major challenge was revealed: people struggled even to express their own conceptualizations and characterizations of some AI systems. Let us consider the following scenario:

> *Upon a user opening a chat window, an intelligent chat-bot shows a recommended set of actions even before the user has entered any text.*

Is this a conversational system or a recommendation system? Is the AI feature initiated by the user (by opening the chat window) or by the system? Our findings suggest that answers to these questions may vary widely even among industry experts in AI product design. The lack of precision and consistency poses challenges for developing a tool like the AI Playbook, but, more importantly, could lead to potential misunderstandings and confusion among practitioners. We thus envision future work in standardizing the language used to discuss concepts that arise at the intersection of interaction design and machine learning.

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6.4 Expanding Accountability

Finally, though the AI Playbook provided concrete actionable guidance, our participants identified the benefits of being even more forceful, holding designers and developers accountable for addressing specific scenarios. Specifically, it was suggested that the AI Playbook be integrated into compliance checkers and task management systems that already gate the distribution of code within the company. Alternatively, the Playbook report could be used as a checklist, where issues are prioritized and addressed prior to moving a design from one phase of development to the next. This raises the additional interesting possibility of dynamically adapting to specific phases of development.

7 CONCLUSION AND ACKNOWLEDGMENTS

Despite the promise of AI to improve the user experience of language-based AI products, errors will remain an inevitable byproduct of the inclusion of AI features. While it is important to prevent or eliminate failures in constructing machine-learned models, practitioners need tools and techniques to plan for unexpected failures in early prototyping and testing of AI user experiences.

To this end, our paper contributes an empirical understanding of the existing prototyping practices of teams developing intelligent language-based software products, as well as the practical challenges that product teams face in so doing. Based on these findings, we present a taxonomy of common errors, and an interactive tool—the AI Playbook—that provide practical guidance on which error scenarios to consider in their designs, and how to simulate such errors in their prototypes.

Through the AI Playbook, our research demonstrates the potential to minimize technical debt by surfacing non-ideal scenarios early in the design process. The benefits of such tool extend beyond prototyping and testing and shows promising returns in design practice, such as promoting standardization and collaboration across disciplines.

We thank all of our study participants and researchers who contributed to our well-rounded understanding of natural language failures and their implications for prototyping practice. We believe that exciting opportunities await HCI researchers and practitioners to expand our knowledge and articulation of AI failures.

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